FACULTY OF BUILT ENVIRONMENT AND SURVEYING

SEMESTER 1

2025/2026

SBEU3213 FIELD ASTRONOMY

CASE STUDY 4 - SEASONAL CHANGES IN SUNRISE DIRECTION

SECTION EB1

LECTURER: DR. AIMI BINTI MUSA & DR. OTHMAN BIN ZAINON

Name	**Matric Number**
MUHAMMAD AFIQ DANIAL BIN ROZAIDIE	A23CS0117
IMAN ABADI BIN MOHD NIZWAN	A23CS0084
MOHAMED ALIF FATHI BIN ABDUL LATIF	A23CS0112
MUHAMMAD AFIQ DANISH BIN MOHD HAZNI	A23CS0118

We declare that this case study is our own work and that all group members contributed to the completion of this assignment.

Group Contributions Statements

1. **MUHAMMAD AFIQ DANIAL BIN ROZAIDIE - A23CS0117**
    a. Contribute on making an introduction.
    b. Contribute on explaining the issue of the case study.
    c. Contribute to summarizing the whole case study investigation.
2. **IMAN ABADI BIN MOHD NIZWAN - A23CS0084**
    a. Contributed to explaining the solar azimuth angle.
    b. Contributed on investigating solar azimuth angle using Stellarium, visualizing how sunrise azimuth angle changes throughout the year.
3. **MOHAMED ALIF FATHI BIN ABDUL LATIF - A23CS0112**
    a. Contributed to explaining the solstices and equinoxes.
    b. Contributed to investigating solstices and equinoxes using EarthSpaceLab to see how tilted earth affects sunrise.
4. **MUHAMMAD AFIQ DANISH BIN MOHD HAZNI - A23CS0118**
    a. Analysed the investigation results and interpreted the scientific meaning behind the changes in sunrise direction.
    b. Developed the analysis and discussion part by transforming the findings into clear explanations.
    c. Addressed common cultural misconceptions and discussed its impacts on daily life, society, education and culture.

Table of Contents

1.0 Introduction

Many people believe that the sun always rises exactly in the east and sets exactly in the west. This belief is often taught at a basic level of education, and that will make some of the people believe it that way until proven wrong. However, if people carefully observe when the Sun will rise over weeks or months, it becomes clear that the Sun's rising point on the horizon is different throughout the year. Additionally, these changes also have a direct effect on the horizon points of the Sun during sunsets. These shifts may be subtle when observed on a daily basis, but they become significant when comparing each point with different seasons.

The seasonal change in sunrise direction is an important astronomical phenomenon that will explain the relationship between the motion of the Earth and the apparent movement of the Sun in the sky. To understand this change on points of the horizon requires knowledge of solar azimuth and the timing of solstices and equinoxes and includes some concepts on the Earth's axial tilt. By studying this case, most people can have a better understanding of why the Sun has a variety of paths during the year and why the idea that the Sun always rises in the east is a very common misconception rather than a scientific fact.

2.0 Case Study Issue

The situation observed in this case study is that the Sun does not always rise in the same direction every day. Instead of always rising exactly in the east every day, the sunrise point will shift along the horizon over the seasons of the year. This change often causes confusion, especially among the individuals who expect the Sun's rising direction to remain constant based on the basic level of knowledge. Figure 2.1 illustrates the seasonal changes of sunrise and sunset direction.

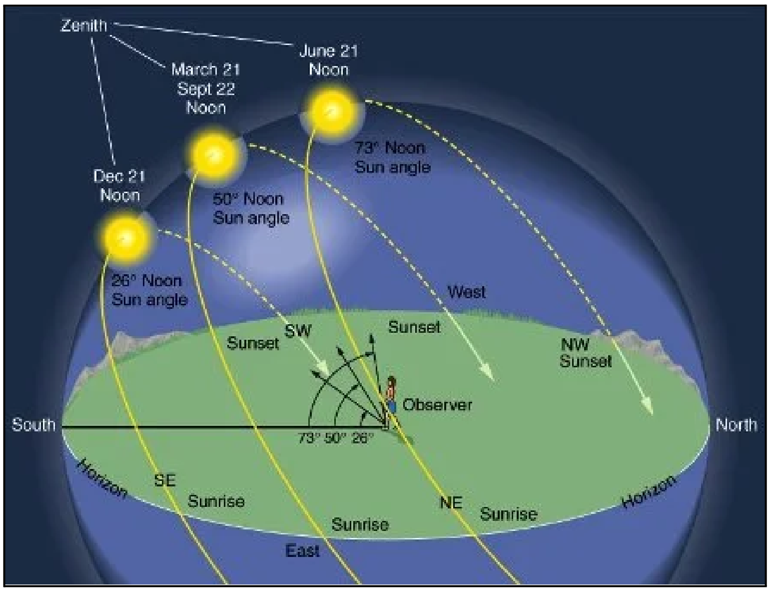

Figure 2.1 Diagram of annual sunrise azimuth changes

This phenomenon occurs in most locations across the world and may become more apparent at the higher latitudes. The change in sunrise direction is the main factor during different seasons, particularly around the solstices and equinoxes. This issue is directly related to astronomy because this phenomenon is caused by the tilt of Earth's axis and its revolution around the Sun. These factors influence the Sun's apparent path across the sky, which is known by solar azimuth. Further details will be investigated throughout this case study.

3.0 Key Astronomical Ideas

3.1 Solar Azimuth

The azimuth angle is an angular measurement that defines the horizontal direction of a celestial object, such as the Sun, as seen from Earth. It is measured along the horizon, commonly beginning at true north and increasing clockwise through east, south, and west, from 0° to 360° (Launch Pad Astronomy, 2017).

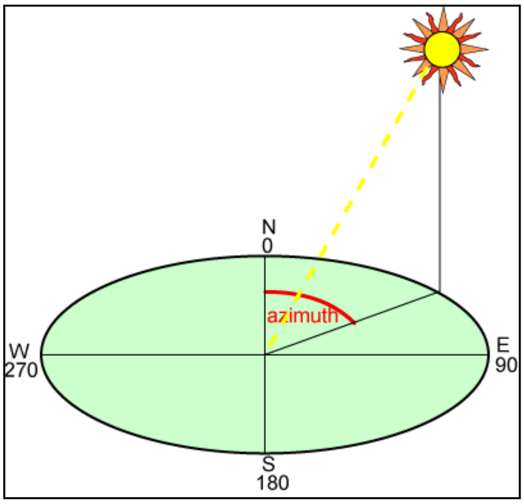

Figure 3.1 Example Azimuth Angle of the Sun (Azimuth Angle | PVEDucation, n.d.)

In the context of sunrise, the azimuth angle indicates the exact compass direction from which the sun appears to rise. For example, an azimuth angle of 90° corresponds to due east, while values less than 90° indicate the Sun rising north of east, and values greater than 90° indicate the Sun rising south of east.

The azimuth angle of the sun at sunrise varies throughout the year due to the Earth's axial tilt and movement around the sun. As Earth circles the sun, the sun's apparent path across the sky varies northward and southward, causing the dawn azimuth to change every day. On the equinoxes, the Sun rises perfectly due east, where the azimuth angle is 90°, whereas on the solstices, the sunrise azimuth deviates the furthest from east.

3.2 Solstices and Equinoxes

The solstices and equinoxes are determined by earth's position in its orbit around the sun. They occur because the earth's axis of rotation is at an angle (23.5 degrees) to the plane on which it orbits the sun (Solstices and Equinoxes, 2017).

Equinoxes

Equinoxes occur twice per year, in March and September. On these days, the Earth is not tilted toward or away from the sun. Because of this, the sun appears straight above the equator. For most areas on earth, the sun rises almost exactly due east and sets due west. During an equinox, the lengths of day and night are almost identical.

Solstices

Solstices occur when the sun is at its most northern or southern point in the sky. The June solstice occurs when the sun rises at its furthest point north of East, resulting in the longest day of the year in the Northern Hemisphere. The December solstice occurs when the sun rises at its farthest point south of the east, resulting in the shortest day of the year. These two solstices mark the extreme dawn positions on the horizon.

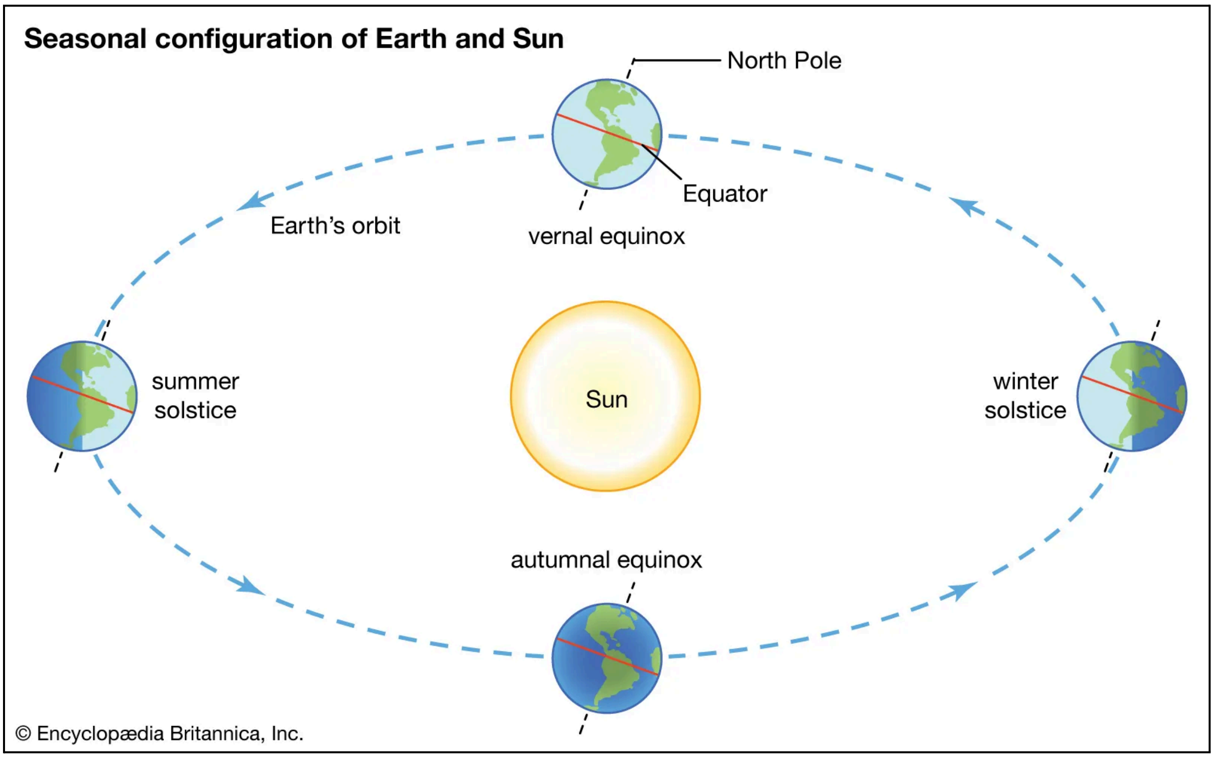

Figure 3.1 Solstices and Equinoxes Each Seasons

Relationship to Latitude

The amount of shift in sunrise direction varies according to one's location on Earth. The sun rises close to the east all year near the equator, therefore the difference is little. The Sun's rising position in mid-latitude regions such as Malaysia varies dramatically north and south over the year. Near the poles, the change is severe, resulting in months of daylight or darkness.

4.0 Investigation

4.1 Solar Azimuth

A sky simulation using Stellarium was used to investigate the sunrise solar azimuth. The observer location was fixed, and sunrise azimuth values were recorded for key dates throughout the year. The simulation showed that the Sun rises due east only on the equinoxes, while on the solstices the sunrise azimuth shifts northward or southward. This demonstrates that sunrise direction varies seasonally due to changes in solar azimuth. The year 2025 was used to simulate the seasonal changes with Johor Bahru, Malaysia, as the location. Figure 4.1 to Figure 4.4 shows the simulation at different times throughout the year.

Figure 4.1 March Equinox Sunrise Simulation

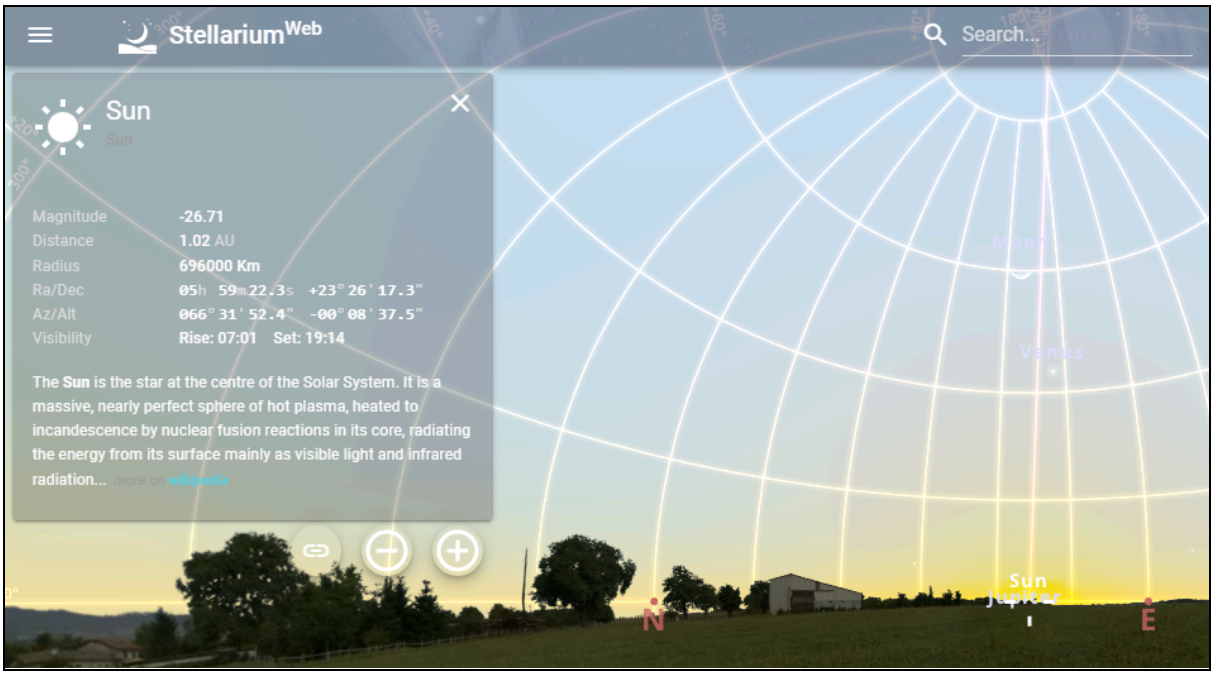

Figure 4.2 June Solstice Sunrise Simulation

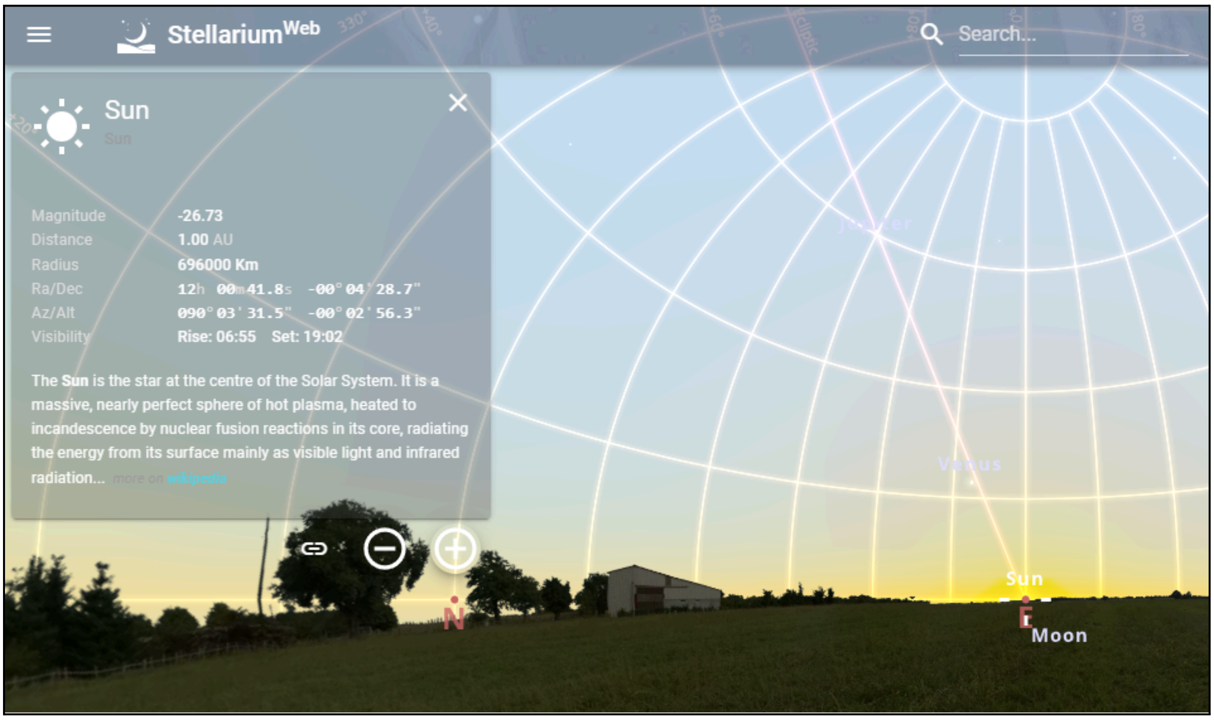

Figure 4.3 September Equinox Sunrise

Figure 4.4 December Solstice Sunrise Simulation

The readings are then recorded in Table 4.1 below, specifying the dates used, the event on the given date, the sunrise azimuth angle, and the direction of sunrise.

Table 4.1 Results of Sunrise Simulation

Date	Event	Sunrise Azimuth (°)	Direction
21/03/2025	March Equinox	89° = ~90°	Due East
21/06/2025	June Solstice	66° < 90°	North of East
23/09/2025	September Equinox	90°	Due East
21/12/2025	December Solstice	113° > 90°	South of East

The azimuth angle of the sunrise changes and shifts from the March equinox until the June solstice, increasing gradually to the north. From the June solstice onwards, the azimuth shifts back to east, where the sun rises at exactly east again during the September equinox. Then, the azimuth angle shifts towards the south, increasing gradually until the December solstice, where it will shift back to the east for another March equinox of the following year, with the same pattern onwards.

4.2 Solstices and Equinoxes

This study uses the EarthSpaceLab Seasons simulation to investigate how the Earth's axial tilt and revolving around the sun affect the direction of sunrise. The Earth was observed at four major points in its orbit, which are the March equinox, the June solstice, the September equinox, and the December solstice. For each position, the earth's tilted axis and the direction of incoming sunlight were investigated. The sun's position relative to the east direction was noted. Screenshots from the simulation were collected to show how the earth's axial tilt causes the sun's apparent path to shift north or south over the year, resulting in different sunrise directions at the equinoxes and solstices.

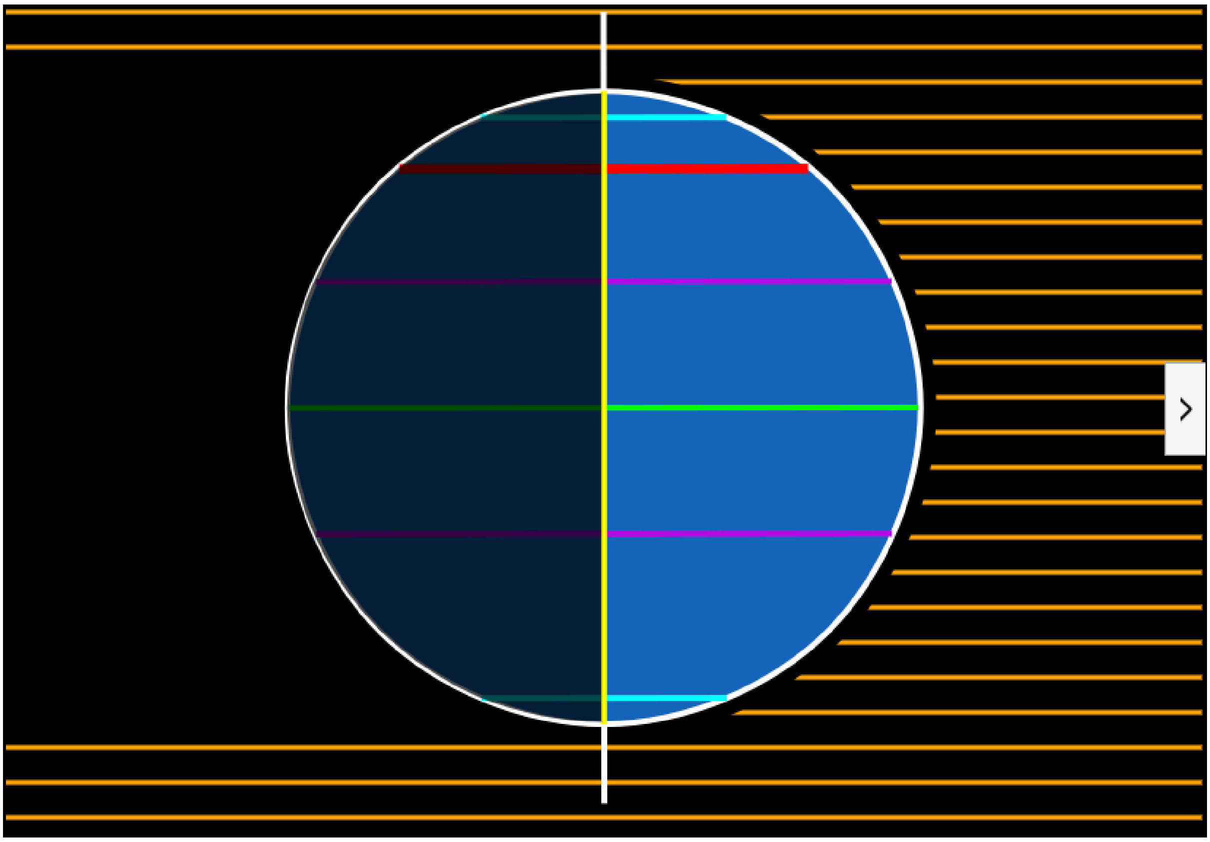

Figure 4.5 March Equinox Earth Axis

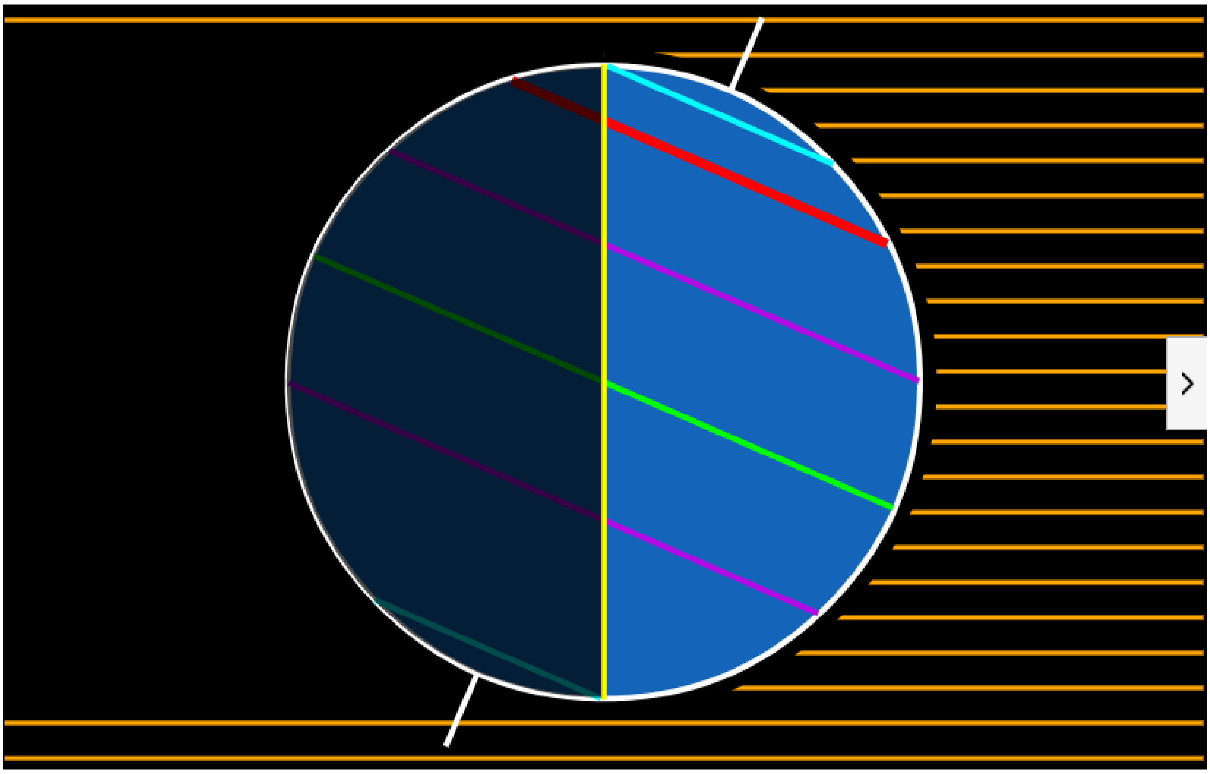

Figure 4.6 June Solstice Earth Axis

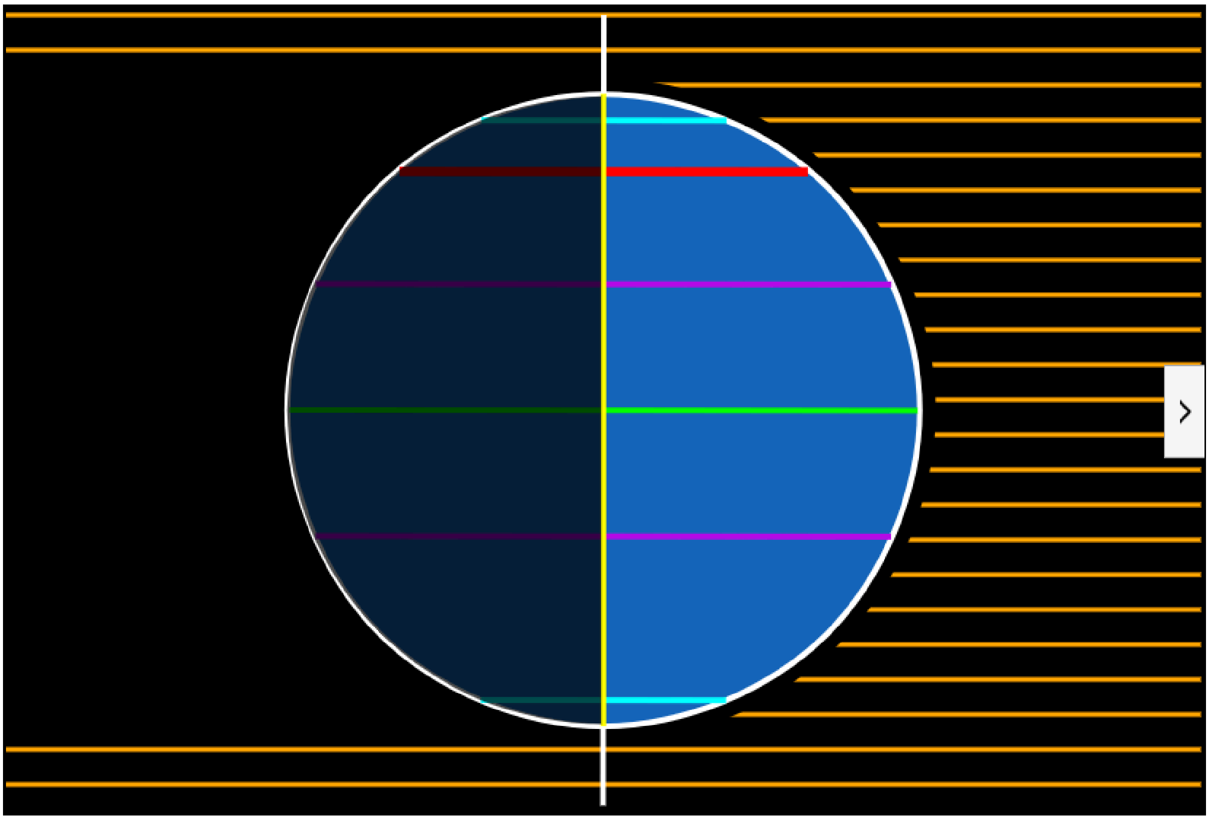

Figure 4.7 September Equinox Earth Axis

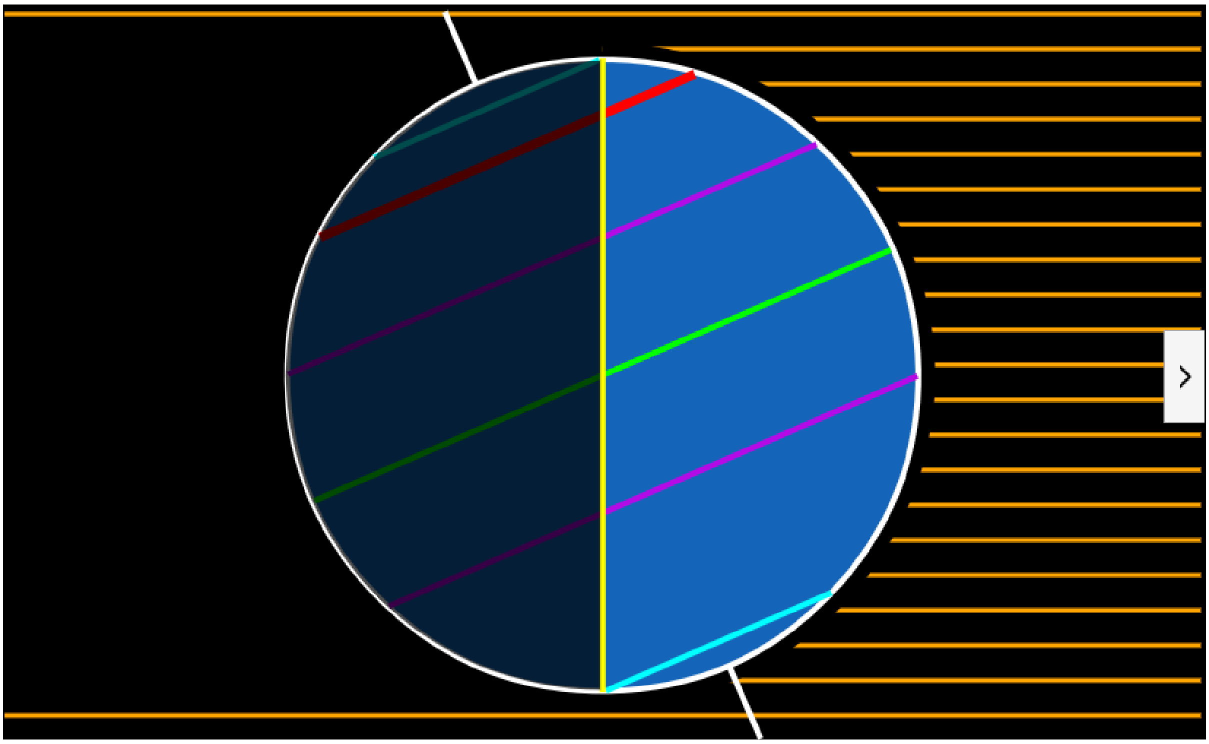

Figure 4.8 December Solstice Earth Axis

The direction of sunrise changes throughout the year as the earth tilts on its axis and orbits the sun. The earth's axis is tilted by approximately 23.5°, and this tilt always points in the same direction in space as the earth travels around its orbit. As a result, when viewed from a fixed point on earth, the sun appears to move in different directions across the sky at different times of the year.

During the March and September equinoxes, the Earth's axis is neither tilted toward nor away from the sun. In this position, the sun is directly above the equator, rising exactly in the east and setting exactly in the west. This is why in both equinox screenshots, the dawn is aligned with the east point on the horizon.

During the June solstice, the Northern Hemisphere tilts toward the sun. This shifts the sun's apparent course northward, causing the sun to rise north of east. In contrast, at the December solstice, the Northern Hemisphere is tilted away from the sun. This causes the sun to rise south of east. The four screenshots clearly demonstrate these variations by showing the earth's tilted axis and the changing direction of incoming sunlight.

5.0 Analysis and Discussion

5.1 Scientific Explanation of the Phenomenon

The seasonal change in sunrise direction is a direct result of the Earth's axial tilt and its revolution around the sun. The Earth is tilted by approximately 23.5° and this tilt remains fixed in orientation as the Earth orbits the sun. Because of this, the apparent path of the sun across the sky shifts northward and southward throughout the year. As demonstrated in the Stellarium simulation for Johor Bahru, the sunrise azimuth is not constant. It is approximately 90° due east only during the equinoxes, shifts north of east during the June solstice, and moves south of east during the December solstice.

This phenomenon can be scientifically explained using the concept of solar azimuth. Azimuth measures the horizontal direction of a celestial object from an observer's position. As the Earth moves around the sun, the sun's declination changes relative to the celestial equator. During the March and September equinoxes, the sun is positioned directly above the equator, causing it to rise exactly in the east. However, during the solstices, the sun reaches its maximum northward or southward displacement. This alters the angle at which the sun's path intersects the horizon, causing the sunrise point to shift along the horizon.

The recorded values from the investigation clearly show this systematic movement: about 66° during the June solstice (north of east) and about 113° during the December solstice (south of east). These changes are gradual and predictable, which forms a yearly cycle. Therefore, the variation in sunrise direction is not random but a natural consequence of the Earth's tilted axis combined with its orbital motion around the sun.

5.2 Cultural Misconceptions

A widespread cultural misconception is the belief that the sun always rises exactly in the east and sets exactly in the west. This idea is commonly taught at an early educational level as a simplified fact and over time it became accepted as an absolute rule rather than an approximation. Because the daily change in the Sun's rising position is very small and difficult to notice without long-term observation, most people never realize that the sunrise point shifts along the horizon throughout the year. As a result, the notion of a fixed east remains deeply rooted in everyday life.

Daily Life

In daily life, this misunderstanding can affect orientation and navigation, especially for individuals who rely on natural cues. Hikers, fishermen, farmers and people in rural areas often use the sun as a reference for direction. Assuming that the sun always rises in the east may lead to errors, particularly during months when the sunrise point is significantly north or south of east. Even in urban settings, people sometimes use the sun's position to estimate direction or time, and this assumption can become misleading when seasonal changes are not fully understood.

Society

At the societal level, this misconception influences how people interpret seasonal changes in daylight. Many are surprised by the shifting times of sunrise and sunset across the year and often attribute these changes only to longer or shorter days without understanding the underlying astronomical reason. Recognizing that the sun's path itself changes helps explain variations in daily routines, working hours and energy usage, especially in regions where daylight duration strongly affects lifestyle and productivity.

Education

In education, oversimplified explanations can limit students' understanding of natural phenomena. Teaching that the sun rises in the east without clarification may create rigid mental models that are difficult to correct later. By encouraging students to observe the horizon over time or use sky simulations, educators can demonstrate that the natural world is dynamic. This approach promotes scientific thinking, curiosity and evidence-based understanding of everyday phenomena.

Environment and Culture

From an environmental and cultural perspective, the changing path of the sun has long influenced human traditions, architecture, and calendars. Ancient civilizations aligned monuments and temples with sunrise positions on solstices and equinoxes, which reflects the early awareness of seasonal solar motion. In modern contexts, building orientation and urban planning consider the sun’s seasonal position to improve thermal comfort and energy efficiency. In tropical regions such as Malaysia, understanding sunlight direction is particularly important for sustainable design and environmental management

6.0 Conclusion and Recommendation

In a nutshell, this case study provides solid proof that the sun does not always rise at a fixed point on the horizon throughout the year but instead either shifts northward or southward depending on the season. Through the use of solar azimuth concepts and sky simulation, it clearly shows that the Sun actually rises exactly due east only during the March and September equinoxes. During the June solstice, the sunrise occurs north of east, while during December solstices, it occurs south of east. These phenomena are caused by the Earth's axial tilt of approximately 23.5° and also its revolution around the Sun, which affects the Sun's apparent path across the sky. The investigation confirms that the common belief that the Sun always rises in the east turns out to be a misconception rather than a scientific fact.

To improve public misunderstanding about the phenomenon, a greater approach should be placed on observational learning and the use of astronomical simulations in education. Schools and universities can encourage students to observe positions daily or just use tools such as Stellarium, SunEarthTools, EarthSpaceLab and SunCalc to visualize seasonal changes. Public awareness can also be enhanced through educational media, social media, and science programs that deliver the explanation that Earth's motion affects daily celestial observations. Practically, understanding seasonal sunrise direction is important for navigation, environmental planning, sustainable building design, religious aspects, and energy efficiency, especially in tropical regions that are closer to the equator, like Malaysia. By replacing oversimplified explanations with clear, evidence-based understanding, people can develop a feeling of appreciation for how astronomical phenomena can influence someone's daily life.

7.0 References

Azimuth Angle | PVEDucation. (n.d.). https://www.pveducation.org/pvcdrom/properties-of-sunlight/azimuth-angle

Launch Pad Astronomy. (2017, September 15). The Sky Part 1: Local Sky and Alt-AZ / Horizon coordinates [Video]. YouTube. https://www.youtube.com/watch?v=i2e0aRtwsCY

Solstices and equinoxes. (2017, June 21). RMetS. https://www.rmets.org/metmatters/solstices-and-equinoxes